

The AI Chip Industry - Final Paper

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Abstract

This paper analyzes the AI Chip industry, specifically Nvidia, which dominates the market. We argue the industry is best characterized as a Stackelberg oligopoly with Nvidia as the leading firm, supported by key suppliers such as Taiwan Semiconductor Manufacturing Company and ASML. High barriers to entry, supply chain constraints, and patented software reinforce this dominance. While innovation improves welfare through quality gains, price markups hinder it. We also examine how policy, including the CHIPS Act, further entrenches industry concentration and shapes global competition.

Introduction

In recent years, the AI chip industry has experienced record breaking growth, and despite ongoing speculation and uncertainty, it is clear that the decisions made by firms and governments today have the potential to reshape the future world economy. From healthcare and transportation to finance and entertainment, advances in artificial intelligence are transforming production processes, increasing productivity, and changing how businesses operate across a wide range of sectors. As a result, geopolitical and economic power is becoming increasingly tied to the development and deployment of AI technologies, and the firms at the forefront of this transformation are those supplying the chips. As the demand for powerful and efficient AI chips continues to grow, understanding the industry and the implications of market power is crucial for policymakers, businesses, and consumers alike.

Throughout this paper, we will reference companies along the supply chain, including but not limited to: Nvidia, Taiwan Semiconductor Manufacturing Company (TSMC), and ASML. The interactions between these companies and their influence on the market structure will be a central focus of this research. Additionally, on the demand side, we will reference companies commonly referred to as “hyperscalers,” which operate large datacenters and make up the primary consumer for AI chips. Taken together, the relationship between these sectors reveal an industry defined by high barriers to entry, strategic interdependence, and concentration.

Overall, this paper argues that the market is best characterized as a Stackelberg oligopoly, with Nvidia acting as the dominant leader in the design of AI accelerators. This market structure grants Nvidia substantial market power, with important implications for prices, welfare, innovation, and competition throughout the broader economy. As such, Nvidia, and its relationship with key players throughout the industry, will be the primary focus of the paper. The paper proceeds as follows: first, it examines the structure of the AI chip industry and its key participants; second, it evaluates how this market aligns with or departs from traditional market structure models; third, it explores the effects of market power on welfare, prices, and entry; and finally, it analyzes recent interactions between firms and governments to assess how policy may shape these outcomes going forward.

Describing the Market & Key Players

Founded in 1993, Nvidia began their journey by inventing the modern graphics processing unit (GPU). Unlike the similarly named central processing unit (CPU), GPUs are designed to perform complex computational tasks simultaneously through a method called parallel processing. While this process, along with Nvidia's later development of CUDA, would be particularly beneficial for computer graphics and video games, researchers quickly realized the broader usefulness of GPUs in other computational heavy tasks, such as machine learning and large language models (LLM).^[1] From that point forward, Nvidia would invest early into programmable and general-purpose GPUs, strategically positioning themselves at the center of AI development and research. Although it would take another decade for the broader commercial potential of AI to fully materialize, the foundations of Nvidia's leadership in AI chips was established during this earlier period. Recognizing Nvidia's role in shaping the technological foundations of modern AI is essential to understanding how the company has secured and maintained its dominant position in the industry. This topic will be further analyzed when we discuss the barriers to entry.

Supply Chain

Although Nvidia can be characterized as the dominant leader in the design and sale of AI accelerators, they do not manufacture these chips themselves. Instead, along the lines of vertical specialization, they deliberately outsource the entirety of the production to third parties. Among these companies the production structure is best characterized as a highly concentrated vertical supply chain, with Nvidia at the top as the leading designer of AI chips, followed by TSMC, which produces the chips on Nvidia's behalf, and then ASML, which provides the equipment necessary for production. Importantly, each layer of the supply chain holds substantial market power of its own, with Nvidia and TSMC's market share estimated to be between 80-90 percent of the AI chip market, while ASML effectively holds a monopoly in the photolithography equipment market.^[2] Although Nvidia faces competition from firms such as AMD and Google, and TSMC faces competition from Samsung and Intel; Nvidia and TSMC's early success in the AI chip industry has allowed them to jump ahead of the competition and establish themselves as early and dominating leaders in the market.

The highly concentrated interdependent relationship between these firms does offer several benefits to the firms involved. Trust mechanisms and information transparency are often strengthened and result in lower transaction costs between firms. In addition to this, firms can more accurately forecast market demand fluctuations and collectively respond to shortages.^[3] However, while this concentrated supply structure can prove more efficient in some regards, market concentration along each level of the supply chain raises valid concerns regarding the possibility of anti-competitive behavior. In addition, there are increased risks of bottlenecks and a phenomenon known as "double marginalization," which occurs when multiple firms with significant market power add a markup to prices throughout the supply

chain, leading to higher prices for consumers and reduced welfare. These topics will be discussed further when we analyze welfare implications later in the paper.

Demand Side

Finally, it is important to establish the consumer side of the market, which is primarily made up of hyperscalers who operate large data centers, such as Google Cloud, Amazon Web Services (AWS), Microsoft Azure, and Oracle. In recent years, these firms have made significant investments in AI and have become the primary consumers of Nvidia's AI chips. These companies either host or offer AI services of their own to their customers, and the demand for AI chips is driven by the need to power these services. There are other consumers of AI chips, such as research institutions and government agencies, however, hyperscalers are the primary drivers of demand in the industry. For instance, Nvidia's most recent earnings report revealed that the datacenter sector accounted for \$62.3 billion in revenue for the fourth quarter, making up about 91% of the company's sales.^[4] Note that this includes all of Nvidia's products, not just AI accelerators. This highlights the significant role that hyperscalers play in the AI chip industry and their influence.

Market Structure

As alluded to earlier, the AI chip industry is best described as a highly concentrated Stackelberg oligopoly, with Nvidia acting as the dominant and leading firm in the design and development of AI chips. In this framework, Nvidia behaves as the first mover, setting the pace of innovation, production, and market expectations, while rival firms, such as AMD, Intel, and other entrants respond to the conditions established by the leader. This first mover advantage allows Nvidia the ability to capture a disproportionate share of profit relative to its competitors. Evidence of this dominance can be seen in annual financial earnings reports. For fiscal year 2025, Nvidia reported \$215.9 billion in revenue with \$193.7 billion coming from datacenters alone.^[4] This represented an estimated 68 percent growth from the previous year, alongside a gross margin of 75 percent. By comparison, datacenter revenue generated by AMD and Intel was an estimated \$16.6 billion and \$16.9 billion, respectively.^[5] While not reflecting overall profits, these earnings highlight the substantial revenue gap between the leader and follower in the Stackelberg oligopoly. For the remainder of this section we elaborate on how Nvidia has captured this position and what this implies for the broader industry.

In the previous section we briefly touched on the history of Nvidia and its role in the development of modern AI models, however the true impact should not be understated. Throughout the 2010s, AI researchers frequently relied on Nvidia GPUs for early deep learning and large language models, as the CUDA platform allowed researchers to repurpose GPUs originally optimized for gaming.^[6] At the time competing companies did not offer similar capabilities, which allowed Nvidia to accumulate a large body of academic research,

open-source tools, and technical knowledge centered around its ecosystem. This generated a powerful consumer lock-in effect, as researchers and firms became increasingly dependent on Nvidia's hardware and software stack. After the release of ChatGPT in 2022 and the resulting surge in demand, these switching costs became even more significant. For datacenters, dependence on Nvidia reflected not only familiarity and accessibility, but also the need for highly efficient, scalable, and easily optimizable hardware.

The open-source tools and academic literature surrounding Nvidia's hardware benefited producers of AI services more broadly. Although the supply side of AI chips remains highly concentrated, the services that come as a result are widely available to the public. Given that one can afford the initial upfront costs, although highly technical, producing a large language model is relatively accessible to the general public. As a result, Nvidia benefits not only from direct demand by hyperscalers, but also from a broader developer base, reinforcing the attractiveness of its platform. Incidentally, many firms have also begun adopting their own AI services, shifting the demand for AI chips outward, and further strengthening Nvidia's position as the leader.

Even so, the question remains whether firms can meaningfully challenge Nvidia's leadership. It is important to remember that the industry is quite young and the substantial growth in demand has signaled many competitors to enter the market. Google, for instance, has developed their own AI accelerator, the Tensor Processing Unit (TPU), and is forecasted to capture about 10% of the datacenter market by 2032.^[7] This demand is driven primarily by the fact that Google Gemini, the company's response to ChatGPT, was trained using their inhouse AI accelerator and has shown promising results. In addition to Google, AMD has also brought an AI accelerator to market, with Meta recently investing into a multi-year, multi-generational partnership to implement AMD infrastructure into their data centers.^[8] Additionally, due to the pricing power that Nvidia holds as the leading firm, hyperscalers have also begun developing their own AI accelerator, such as AWS, marketing them as a high performance and low cost alternative. However, despite attempts to reduce dependency on Nvidia, these firms will likely remain dependent on the firm, especially considering that newer datacenters are being built around Nvidia architectures.

Barriers to Entry

The high-end AI GPU market is one of the hardest markets to enter in the world. Barriers to entry operate at multiple levels through capital, software, supply chains, and talent. These mechanisms interact and reinforce each other in ways that make new entry nearly impossible.

The most fundamental barrier is sheer financial scale. The substantial research and development costs, coupled with the complexities of designing and manufacturing advanced, high-end performance chip architectures directly impede new competition entry into the AI chip market. Developing an advanced 3nm GPU chip is estimated to cost between \$500 million and \$1.5 billion, depending on complexity.^[9] Advanced nodes, 3nm and below, like

those produced by Nvidia, require massive investments in design, verification, and software development. This has led to industry consolidation, where only a few companies can afford to compete on the razor's edge of semiconductor innovation.

Once a chip is designed by Nvidia, it must be fabricated on silicon wafers using extreme precision. This is called the fabless model by Nvidia, where they leave the actual chip fabrication up to TSMC. For nodes smaller than 3nm, like Nvidia's chips, TSMC's market share is around 95%, making it the de facto near-monopoly in advanced node fabrication. For any new AI GPU maker, there is essentially one viable supplier for their fabrication process, with foundry access being constrained to TSMC. Unlike any new GPU designer who is waitlisted or priced out, Nvidia has a long standing, priority relationship with TSMC.

However, even before TSMC can make a chip, they have to receive machines from a single company, ASML, who have a one-hundred percent monopoly in creating the advanced chips 7nm and below. The machines that make ASML exclusive are the EUV (Extreme Ultraviolet) lithography machines. They use light waves to carve out the transistor patterns onto silicon wafers. This highly advanced machine has over 100,000 components and weighs 180 tons, and it took 30 years to develop.^[10] ASML, being the only company that produces the EUV lithography machines that make the high end chips, creates a choke point.

Another barrier to entry is talent, which is the most underrated barrier to entry in AI chips because, unlike capital, you can't just buy your way out of it overnight. The supply of engineers possessing the requisite expertise in GPU architecture, parallel computing, and AI-specific hardware design is structurally insufficient relative to industry demand. Evidence suggests the global semiconductor workforce will require more than one million additional skilled workers by 2030, with the existing pipeline falling well short of that figure.^[11] This scarcity constrains potential entrants, who face not only high capital requirements but also binding labor market constraints that limit their capacity to staff competitive R&D operations.

A significant portion of the knowledge underlying Nvidia's product development is accumulated through decades of iterative engineering and is not readily transferable through formal training or labor market recruitment alone. Nvidia's Compute Unified Device Architecture (CUDA) ecosystem, developed over nearly two decades, represents a substantial stock of embedded organizational knowledge. CUDA is the programming model developed by Nvidia to work specifically with Nvidia chips. Engineers with deep familiarity with CUDA's architecture, optimization techniques, and integration with downstream software frameworks constitute a form of firm specific human capital that cannot be rapidly acquired by new workers, even under competitive compensation conditions.

With greater financial resources available for compensation, retention incentives, and internal development programs, Nvidia is able to attract and retain engineers who might otherwise contribute to competing firms or new entrants. This dynamic reflects a form of oligopsonistic influence in the market for specialized AI engineering talent, whereby incumbent profitability translates directly into a labor market advantage that prospective competitors cannot easily overcome.

Products & Pricing

Nvidia does not distribute its advanced AI accelerators through a single transparent pricing mechanism. Rather, the price realized by any given transaction reflects the interaction of multiple structural layers, including hardware production costs, memory supply constraints, software bundling, volume arrangements, competitive dynamics, and regulatory factors. Because this pricing operates largely outside publicly disclosed schedules, the most tractable method for analyzing it is through the resale market, which provides observable price signals and allows one to assess the relative contribution of each layer to the final market price.

The first and foundational layer is Nvidia's own production and distribution model. The primary buyers of Nvidia's high-end accelerators, referred to as hyperscalers, include Amazon Web Services, Microsoft Azure, Google Cloud, Meta, OpenAI, large research institutions, and Fortune 500 enterprises. These entities do not purchase discrete GPU units; rather, they procure integrated systems comprising the GPU hardware alongside NVLink interconnects, software toolkits, server rack infrastructure, and enterprise support. Pricing therefore, operates at the system level rather than the component level, and is determined by factors including purchase volume, prevailing supply scarcity, contract duration, and priority allocation, a mechanism under which buyers compete for access to constrained supply rather than negotiating on price alone.

The second layer is the GPU rental market, in which end users access compute capacity leased from hyperscalers rather than purchasing hardware outright. Rental prices are publicly observable. Nvidia's H100 gpu currently ranges from approximately \$1.49 to \$6.98 per GPU-hour on major cloud platforms.^[12] This market exhibits a form of price discrimination structured around tolerance for service interruption and urgency of compute access. Customers are segmented across what may be characterized as a risk-tolerance continuum. Those at the lower end access capacity only when excess supply exists and are subject to preemption on short notice, while those at the upper end pay premium rates for guaranteed allocation and scheduling priority.

The emergence of AMD as a credible competitor in the high-end accelerator market has introduced a modest but meaningful competitive constraint on Nvidia's pricing, particularly in the rental segment. Using resale markets like eBay, one can deduce prices. AMD's MI300X accelerator carries a list price of approximately \$15,000 while delivering 192 GB of onboard memory, compared to Nvidia's H100 at approximately \$32,000 with 80 GB.^[13]

The persistence of Nvidia's price premium reflects the weight of Nvidia's software ecosystem as a pricing factor independent of hardware specifications. While the MI300X delivers superior memory capacity and demonstrates performance advantages for memory-bound inference tasks. AMD's ROCm software platform continues to lag behind CUDA in stability, tooling maturity, and framework compatibility.

In aggregate, AMD's entry into the high-end accelerator segment has introduced price competition at the hardware and rental levels, but has not yet meaningfully eroded Nvidia's

pricing power in the enterprise and hyperscaler segments where software lock-in, supply priority relationships, and system-level bundling remain the dominant determinants of market price.

Welfare Implications

The topic of welfare in the AI chip industry is expansive, as every firm, government, and country must be considered to get an accurate picture of winners and losers. Although the scope of this paper is limited to the supply chain of AI accelerators, looking outward to the end-user of AI services introduces important dynamics that must also be considered. To capture this, the section begins by establishing the business-to-business welfare between firms along the supply chain and those directly consuming Nvidia AI chips. We then broaden our focus to social welfare and how the broader economy is affected by firm decisions, including any externalities that are unique to this highly concentrated market structure.

Because the market analyzed throughout this paper is a vertical supply chain, we must redefine the standard interpretation of consumer and producer surplus. AI accelerators are best described as intermediate goods used in the production process of AI services. Although there are many use cases for GPUs outside of this production process, we restrict the market to hyperscalers primarily, as this group is currently driving the majority of demand. Welfare, in this case, is then defined as the economic gains from trade for both parties and is represented by the difference between transaction price and what buyers are willing to pay. Framing welfare this way makes it easier to evaluate how Nvidia's pricing, output, and product quality influence the economic gains from trade and how hyperscalers' demand curve shifts in response. In the following section, two main channels are highlighted, a cost channel and a quality channel, which can influence the total welfare of the industry.

Moore's law, a theory stating that every two years the amount of transistors on a semiconductor will double, has been a prevailing force in technological growth over the past century. The amount of transistors a supplier can compact into a single chip, and the architectural framework supporting developers, work in tandem to produce more efficient and faster GPUs. Over the last decade, Nvidia and TSMC have significantly optimized this production process, greatly improving the consumers' use through both quality and cost channels. For reference, Nvidia's Blackwell architecture (2024) has approximately 128 billion more transistors than their previous Hopper architecture (2022), and is about 25x more energy efficient.^[14] The efficiency of a chip is typically measured as the amount of computational work done for every watt of power consumed, also commonly referred to as performance per watt.

Cost and Quality Channels

In terms of the cost channel, innovations by Nvidia benefit hyperscalers through reductions in both marginal and fixed costs. For marginal costs, hyperscalers require a substantial

amount of electricity for both powering and cooling their datacenters. It is estimated that these firms consume roughly 4% of the United States total electricity each year, with this predicted to rise as more datacenters are constructed.^[15] Improvements in efficiency directly lead to lower marginal costs in the form of electricity usage, thus shifting the willingness to pay for Nvidia chips outward, as hyperscaler profits would be expected to increase as a result. As GPUs become more efficient, fixed costs are also impacted, as the space required to house AI accelerators is reduced while maintaining comparable performance to previous architectures. Overall, as a result of this cost channel, the profit that hyperscalers expect is increased, and thus the willingness to pay will shift outward, and increase total welfare for both seller and buyer.

In addition to the cost channel, a quality channel must also be considered. As we mentioned earlier, the final consumer market of this supply chain are the AI services being produced. Firms that produce these services are interested in the latency of the model and the accuracy of the information being relayed. Innovations by Nvidia impact both of these through the amount of data and speed at which each GPU can compute. The industry typically refers to this process as tokenization, which is the process of breaking down prompts by tokens and inferring which tokens to respond with.^[16] It is important to preface that these open-source AI services typically operate on a subscription basis, so the user-experience and productivity benefits is what drives consumers to purchase their product. The speed and accuracy of this tokenization process directly improves user-experience, allowing firms to capture a larger share of profit. Again, similar to the cost channel, this quality channel pushes the willingness to pay outward, and raises total welfare for both parties.

Outside of the mechanisms, we can empirically analyze the economic gains by both seller and buyer through annual financial reports. In 2025, Nvidia captured \$115 billion in revenue from its datacenter segment, with a year over year growth of 114% and a gross margin of 74%.^[17] In comparison, Microsoft Azure ended the year with \$75 billion in revenue, with a year over year growth of 34% and a gross margin of 69%.^[18] On an individual firm basis, we can see that the surplus for both have increased substantially, albeit with Nvidia capturing a greater share. Aggregate market surplus is more difficult to quantify and would require summing the total profit for each hyperscaler. This is difficult for a number of reasons, however the main issue is that hyperscaler revenue is not entirely attributed to Nvidia's product. These firms offer a wide range of services that fall outside the bounds of AI accelerators, and thus should be considered when analyzing these figures. Putting this aside, even if we were generous with the percentage of revenue that hyperscalers earn as a result of Nvidia products, it is clear that surplus is leaning toward the seller at the firm level.

Market Power and Welfare Distribution

Earlier we discussed the market structure of the industry, from the concentrated vertical supply chain to the stackelberg oligopoly, and we see many of those same dynamics play out in how welfare is distributed. Given the market power Nvidia holds, we can observe the firm's

ability to disproportionately capture more profit relative to the buyer through lower output, bottlenecks, and markups along each layer of the supply chain. Recently, due to the immense demand shock for AI accelerators, ram and semiconductor shortages have reduced the overall output of Nvidia GPUs, resulting in upward pressure on prices.^[19]

Although the buyer's willingness to pay has risen due to cost and quality channels, the actual price that firms are paying is also rising, reducing the total surplus that hyperscalers could potentially capture. Nvidia understands the short run gain of these scenarios, however movements by the firm suggest that they have a long run incentive to keep prices relatively affordable, advertently keeping hyperscalers dependent on their full-stack of hardware. As previously mentioned, firms like AWS and Google Cloud have begun investing into their own AI accelerators as a response to Nvidia's pricing power and recapture lost profit margins. Nvidia has partially circumvented this by dealing in long-term multi-generational contracts directly with hyperscalers and AI service companies.

Finally, we briefly touch on social surplus and the costs associated with AI accelerators. The association between AI accelerators and externalities is through more indirect channels, as these costs are typically associated with the final market goods, such as datacenters and AI services. With this in mind, we have observed widespread growth in datacenter capacity buildouts, with an estimated \$7 trillion in investments by 2030.^[20] Accompanying this are important questions regarding renewable energy infrastructure and the negative externalities associated with pollutants, capital accumulation, and labor dynamics. For labor dynamics, despite greater productivity through AI services, the overall impact on labor markets and outcomes remain to be seen. The largest factor determining the impact of AI on labor markets is whether it is complementary or substitutable with specific occupations. However, as we experience greater technological advancements in AI accelerators, only time will tell how obscure the line between the two becomes. Overall, the determining factor for social welfare is how governments react to the growing industry and the regulations they choose to adopt.

Government & Firm Interactions

The enactment of the Creating Helpful Incentives to Produce Semiconductors (CHIPS) and Science Act in August 2022 represents the most significant intervention by the U.S. federal government in the semiconductor industry since the postwar period. The legislation operates through two complementary mechanisms: domestic subsidy provision and extraterritorial export control, both of which have structural implications for barriers to entry in the AI chip market.

The Act authorizes approximately \$280 billion in new funding, of which \$52.7 billion is appropriated directly for semiconductor manufacturing, research, and workforce development. This includes \$39 billion in manufacturing subsidies and a 25% investment tax credit for semiconductor manufacturing equipment, with the dual objectives of

strengthening U.S. supply chain resilience and constraining Chinese technological advancement.^[21] These subsidies disproportionately benefit established U.S. based firms and their manufacturing partners, as the capital requirements for qualifying new fabrication facilities. By early 2024, analysts estimated the Act had incentivized between 25 and 50 separate projects representing projected total investments of \$160 to \$200 billion and 25,000 to 45,000 new jobs.^[22]

A central provision of the Act functions as a geographic constraint on capital allocation. Recipients of CHIPS Act funding are prohibited from expanding semiconductor manufacturing capacity in China or in any country designated as a national security threat for a period of ten years from the date of funding receipt. The Act will unlock financing provided the recipient does not build comparable facilities in China, representing the most significant shift in U.S. technology transfer policy since the 1990s.

The CHIPS Act operated in conjunction with a parallel regulatory framework administered by the Commerce Department's Bureau of Industry and Security (BIS). In October 2022, the administration imposed controls on the export of designated semiconductor types, computer systems, and fabrication equipment to China.^[23] The stated objectives of impairing Chinese AI and supercomputing capabilities. Preventing China from designing and producing its own advanced devices by restricting access to Western design tools and manufacturing equipment, and foreclosing independent Chinese development of semiconductor manufacturing equipment. These controls were subsequently tightened in 2023, 2024, and 2025. BIS closed gaps in controls for certain Nvidia chips designed for China, expanding controls to consider a chip's total processing performance and performance density, and extended restrictions to include Chinese design firms using U.S. technology.

The combined effect of the CHIPS Act subsidy regime and the accompanying export control framework is to segment the global semiconductor market along geopolitical lines. Channeling demand for advanced AI chips toward a small set of U.S. aligned producers. These controls focus primarily on the cutting edge of semiconductor design and manufacturing process and, while effective in slowing the PRC's advancement in the most advanced technology, do not significantly constrain China's ability to continue manufacturing older legacy chips.

Taken together, the CHIPS and Science Act and its accompanying export control framework constitute an aggressive change to the high end chip manufacturing market. By directing capital toward established U.S. based products through subsidies and tax incentives, while simultaneously restricting the access of foreign competitors to advanced manufacturing inputs and end market customers. The legislative and regulatory regime has structurally reinforced the concentration of leading edge AI chip production among a small number of incumbent firms, including Nvidia. This intervention does not merely reflect existing market conditions but strengthens them.

Conclusion

The AI chip industry is a rapidly evolving and highly concentrated market, with Nvidia emerging as an early and dominant leader in the design of AI accelerators. The industry's structure, characterized by its concentrated vertical supply chain and significant barriers to entry, has important implications for market power, pricing, and welfare distribution. We expect to see continued growth in demand for AI chips, driven by the expansion of AI services and applications, which will further solidify Nvidia's position in the market. However, the emergence of competitors like AMD and Google, along with potential regulatory interventions, may introduce new dynamics that could allow for more competition and innovation in the industry. Foundries like TSMC and equipment suppliers like ASML will also play a critical role in shaping the future of the AI chip market, as their control over key production inputs gives them significant influence over the industry's trajectory. Monitoring these developments will be essential for understanding the evolving landscape of the AI chip industry and its broader economic implications.

Overall, the AI chip industry serves as a compelling case study of how technological innovation, market structure, and government policy interact to shape the industry we see today. Global economic growth and the landscape of sectors from healthcare to transportation, and many more, will evolve alongside the development of AI, and thus the firms that produce the underlying hardware will have a tremendous influence on the future of geopolitical power, social welfare, and how we interact with technology on a daily basis. As such, understanding the dynamics of this industry is crucial for policymakers, businesses, and consumers alike as we navigate the rapidly changing technological landscape. To that end, this paper has provided an analysis of the key players, market structure, barriers to entry, pricing dynamics, welfare implications, and government interactions that define the AI chip industry. As the industry continues to evolve, ongoing research and analysis will be necessary to fully understand its implications for the broader economy and society.

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